

```
!pip install matplotlib
```

```
Requirement already satisfied: matplotlib in c:\users\subhasri\
anaconda3\lib\site-packages (3.10.6)
Requirement already satisfied: contourpy>=1.0.1 in c:\users\subhasri\
anaconda3\lib\site-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cyclor>=0.10 in c:\users\subhasri\
anaconda3\lib\site-packages (from matplotlib) (0.11.0)
Requirement already satisfied: fonttools>=4.22.0 in c:\users\subhasri\
anaconda3\lib\site-packages (from matplotlib) (4.60.1)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\subhasri\
anaconda3\lib\site-packages (from matplotlib) (1.4.9)
Requirement already satisfied: numpy>=1.23 in c:\users\subhasri\
anaconda3\lib\site-packages (from matplotlib) (2.3.5)
Requirement already satisfied: packaging>=20.0 in c:\users\subhasri\
appdata\roaming\python\python313\site-packages (from matplotlib)
(26.0)
Requirement already satisfied: pillow>=8 in c:\users\subhasri\
anaconda3\lib\site-packages (from matplotlib) (12.0.0)
Requirement already satisfied: pyparsing>=2.3.1 in c:\users\subhasri\
anaconda3\lib\site-packages (from matplotlib) (3.2.5)
Requirement already satisfied: python-dateutil>=2.7 in c:\users\
subhasri\anaconda3\lib\site-packages (from matplotlib) (2.9.0.post0)
Requirement already satisfied: six>=1.5 in c:\users\subhasri\
anaconda3\lib\site-packages (from python-dateutil>=2.7->matplotlib)
(1.17.0)
```

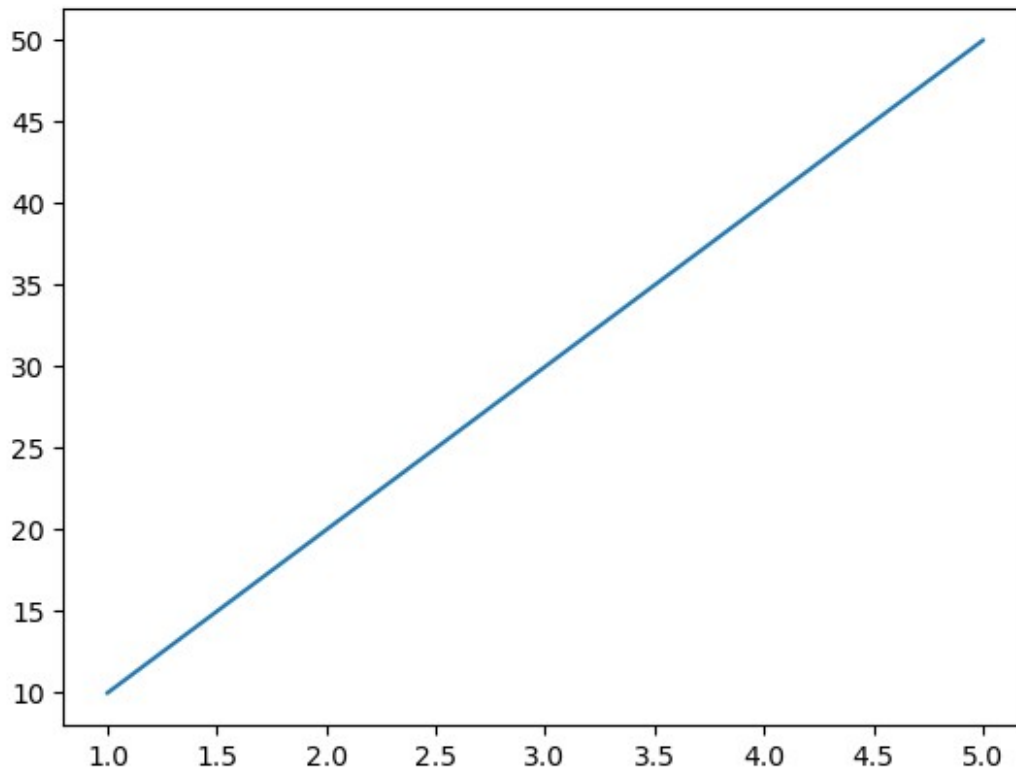
```
import matplotlib.pyplot as plt
```

```
x = [1,2,3,4,5]
```

```
y = [10,20,30,40,50]
```

```
plt.plot(x, y)
```

```
plt.show()
```

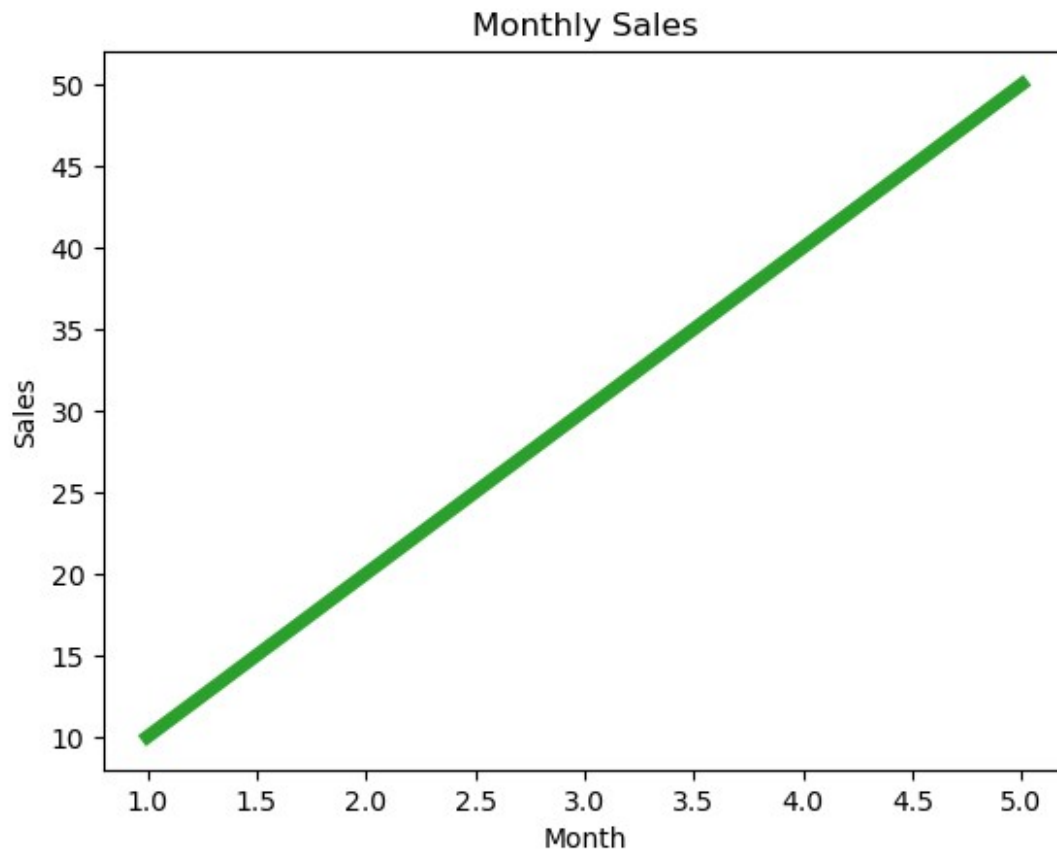


```
import matplotlib.pyplot as plt

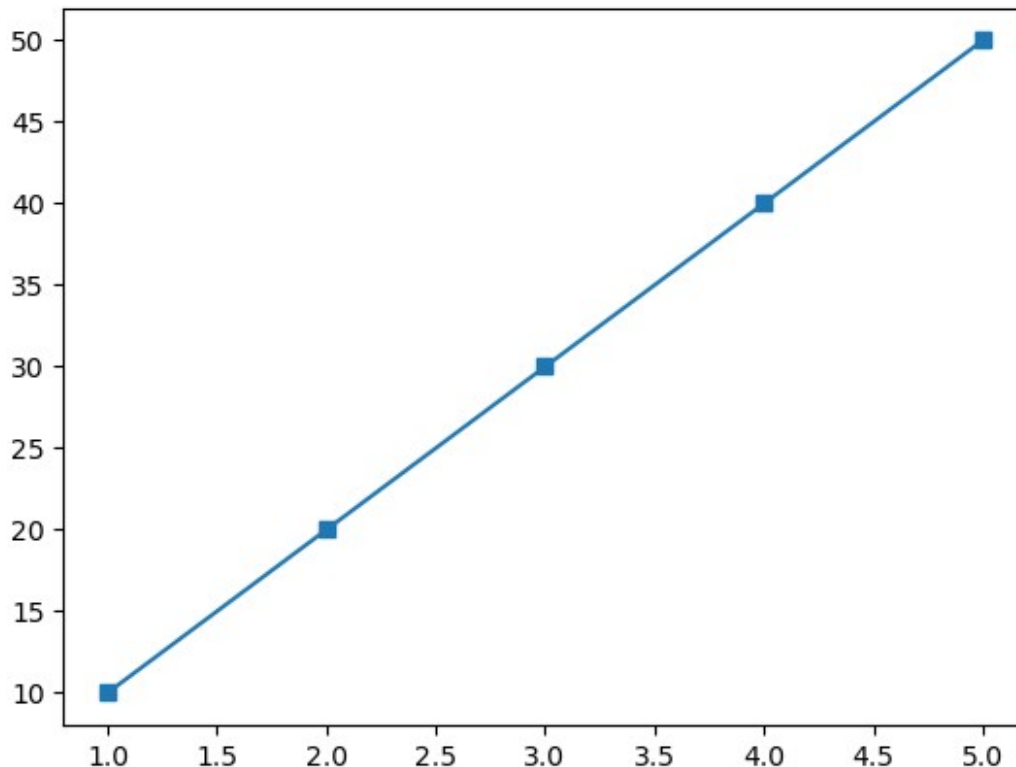
fig, ax = plt.subplots()

ax.plot([1,2,3], [10,20,30,])
plt.title("Monthly Sales")
plt.xlabel("Month")
plt.ylabel("Sales")
plt.plot(x, y, color='red')
plt.plot(x, y, linestyle='-.')
plt.plot(x, y, linewidth=5)

[<matplotlib.lines.Line2D at 0x25dbdedd810>]
```



```
plt.plot(x, y, marker='s')  
[<matplotlib.lines.Line2D at 0x25dbelad1d0>]
```



```
!pip install seaborn
```

```
Requirement already satisfied: seaborn in c:\users\subhasri\anaconda3\lib\site-packages (0.13.2)
```

```
Requirement already satisfied: numpy!=1.24.0,>=1.20 in c:\users\subhasri\anaconda3\lib\site-packages (from seaborn) (2.3.5)
```

```
Requirement already satisfied: pandas>=1.2 in c:\users\subhasri\anaconda3\lib\site-packages (from seaborn) (2.3.3)
```

```
Requirement already satisfied: matplotlib!=3.6.1,>=3.4 in c:\users\subhasri\anaconda3\lib\site-packages (from seaborn) (3.10.6)
```

```
Requirement already satisfied: contourpy>=1.0.1 in c:\users\subhasri\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (1.3.3)
```

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Requirement already satisfied: cycler>=0.10 in c:\users\subhasri\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (0.11.0)
```

```
Requirement already satisfied: fonttools>=4.22.0 in c:\users\subhasri\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (4.60.1)
```

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Requirement already satisfied: packaging>=20.0 in c:\users\subhasri\appdata\roaming\python\python313\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (26.0)
```

Requirement already satisfied: pillow>=8 in c:\users\subhasri\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (12.0.0)
Requirement already satisfied: pyparsing>=2.3.1 in c:\users\subhasri\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (3.2.5)
Requirement already satisfied: python-dateutil>=2.7 in c:\users\subhasri\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in c:\users\subhasri\anaconda3\lib\site-packages (from pandas>=1.2->seaborn) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in c:\users\subhasri\anaconda3\lib\site-packages (from pandas>=1.2->seaborn) (2025.2)
Requirement already satisfied: six>=1.5 in c:\users\subhasri\anaconda3\lib\site-packages (from python-dateutil>=2.7->matplotlib!=3.6.1,>=3.4->seaborn) (1.17.0)

```
import seaborn as sns
import matplotlib.pyplot as plt

x = [1,2,3,4,5]
y = [10,20,30,40,50]

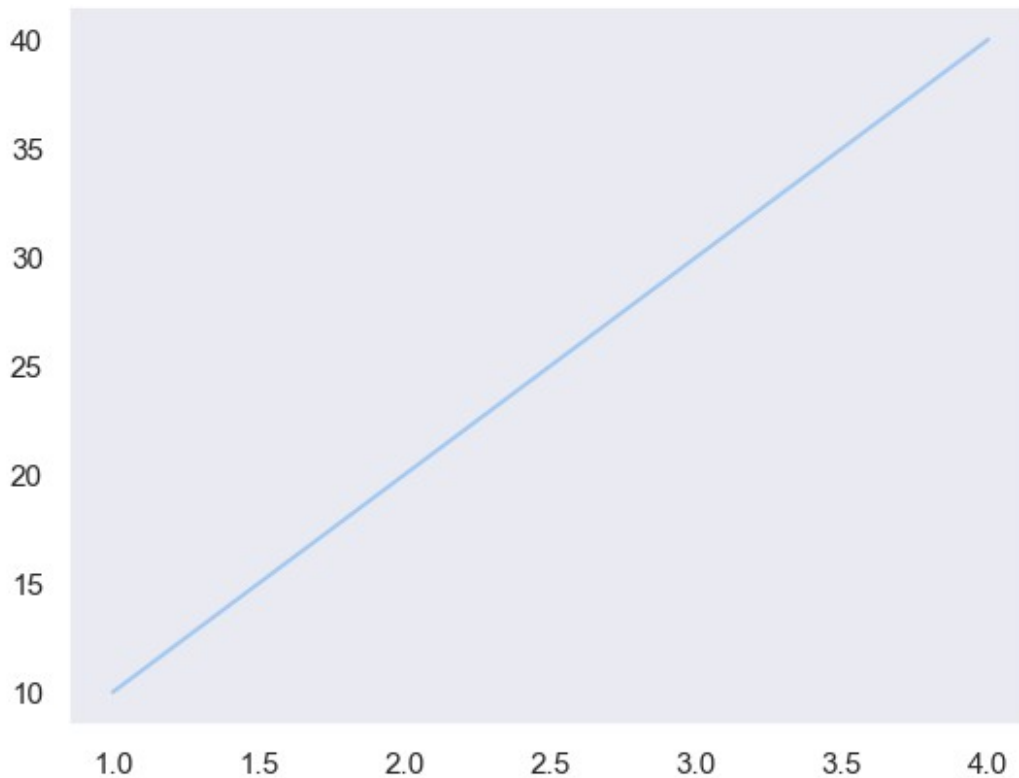
sns.set_theme()

sns.set_style("dark")

sns.set_palette("pastel")

sns.lineplot(
    x=[1,2,3,4],
    y=[10,20,30,40]
)

plt.show()
```



```
print(sns.get_dataset_names())
```

```
['anagrams', 'anscombe', 'attention', 'brain_networks', 'car_crashes',  
'diamonds', 'dots', 'dowjones', 'exercise', 'flights', 'fmri',  
'geyser', 'glue', 'healthexp', 'iris', 'mpg', 'penguins', 'planets',  
'seaice', 'taxis', 'tips', 'titanic']
```

```
import seaborn as sns
```

```
df = sns.load_dataset("titanic")
```

```
print(df.head())
```

survived	pclass	sex	age	sibsp	parch	fare	embarked
0	3	male	22.0	1	0	7.2500	S
1	1	female	38.0	1	0	71.2833	C
2	3	female	26.0	0	0	7.9250	S
3	1	female	35.0	1	0	53.1000	S
4	3	male	35.0	0	0	8.0500	S

	who	adult_male	deck	embark_town	alive	alone
0	man	True	NaN	Southampton	no	False
1	woman	False	C	Cherbourg	yes	False
2	woman	False	NaN	Southampton	yes	True
3	woman	False	C	Southampton	yes	False
4	man	True	NaN	Southampton	no	True

```
import matplotlib.pyplot as plt

months = ['Jan', 'Feb', 'Mar', 'Apr']
sales = [100, 150, 200, 250]

plt.plot(months, sales, marker='o')

plt.title('Monthly Sales')
plt.xlabel('Month')
plt.ylabel('Sales')

plt.show()
```



```
import matplotlib.pyplot as plt

dept = ['HR', 'IT', 'Sales']
```

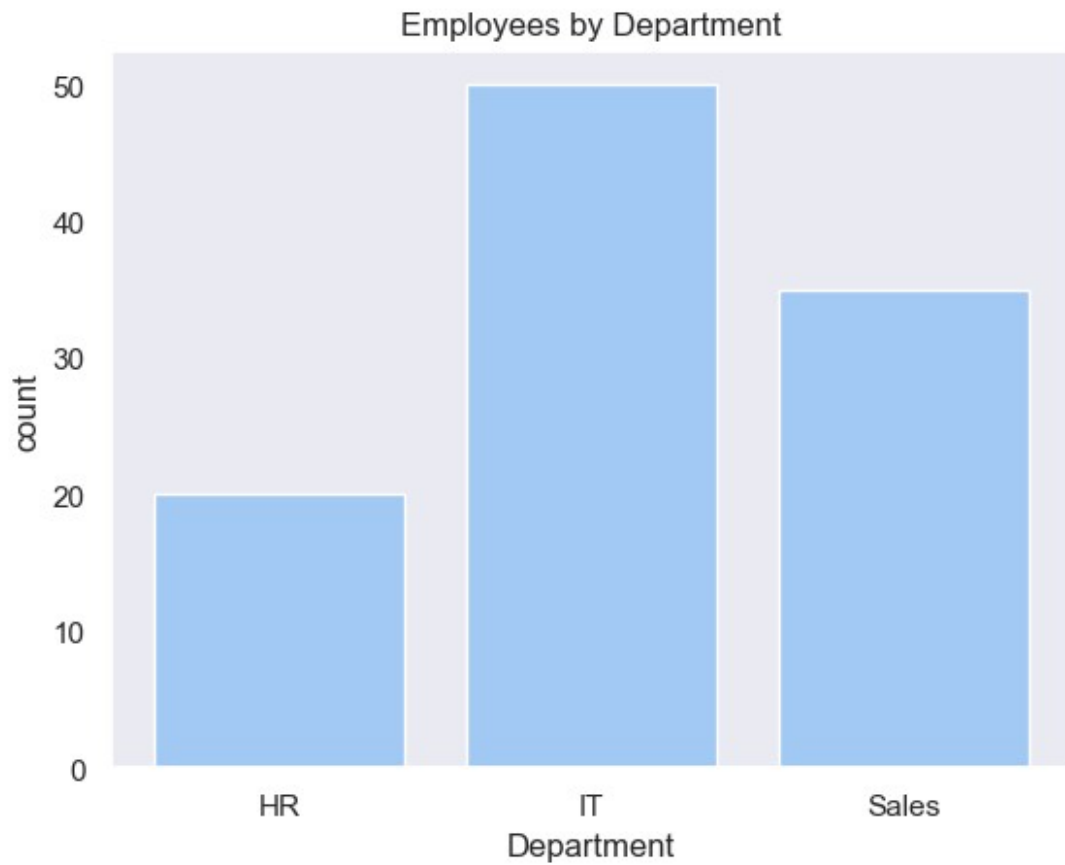
```

employees = [20, 50, 35]
plt.bar(dept, employees)

plt.title("Employees by Department")
plt.xlabel("Department")
plt.ylabel("count")

plt.show()

```



```

import matplotlib.pyplot as plt

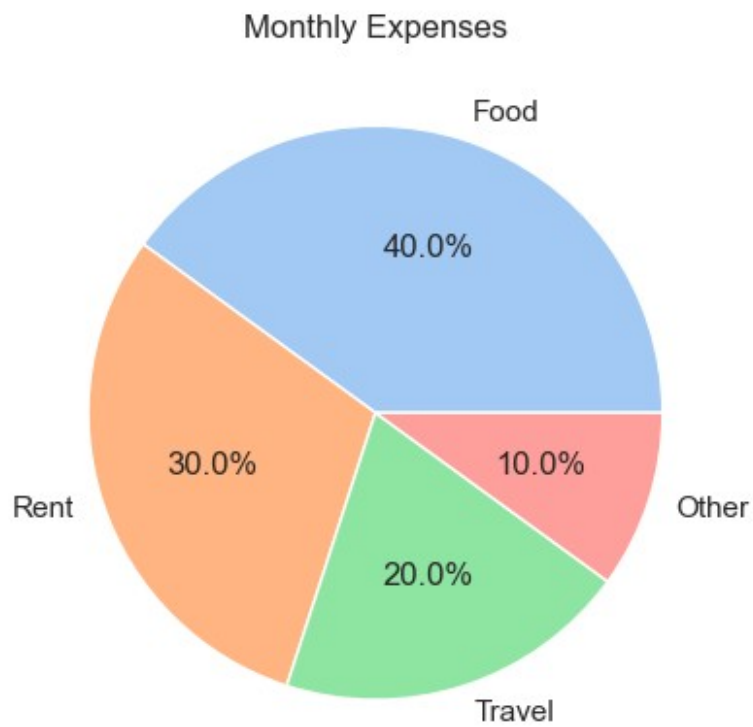
labels = ['Food', 'Rent', 'Travel', 'Other']
values = [40, 30, 20, 10]

plt.pie(
    values,
    labels=labels,
    autopct='%1.1f%%'
)

plt.title("Monthly Expenses")

```

```
plt.show()
```



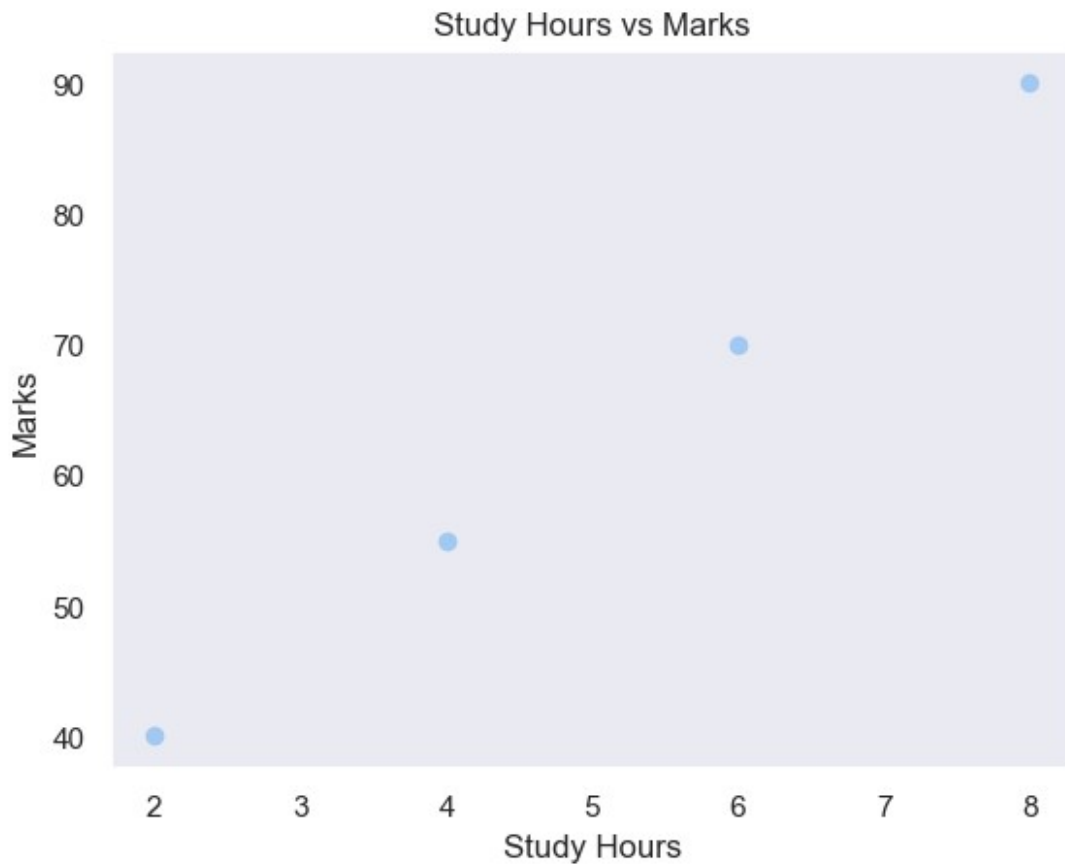
```
import matplotlib.pyplot as plt

hours = [2,4,6,8]
marks = [40,55,70,90]

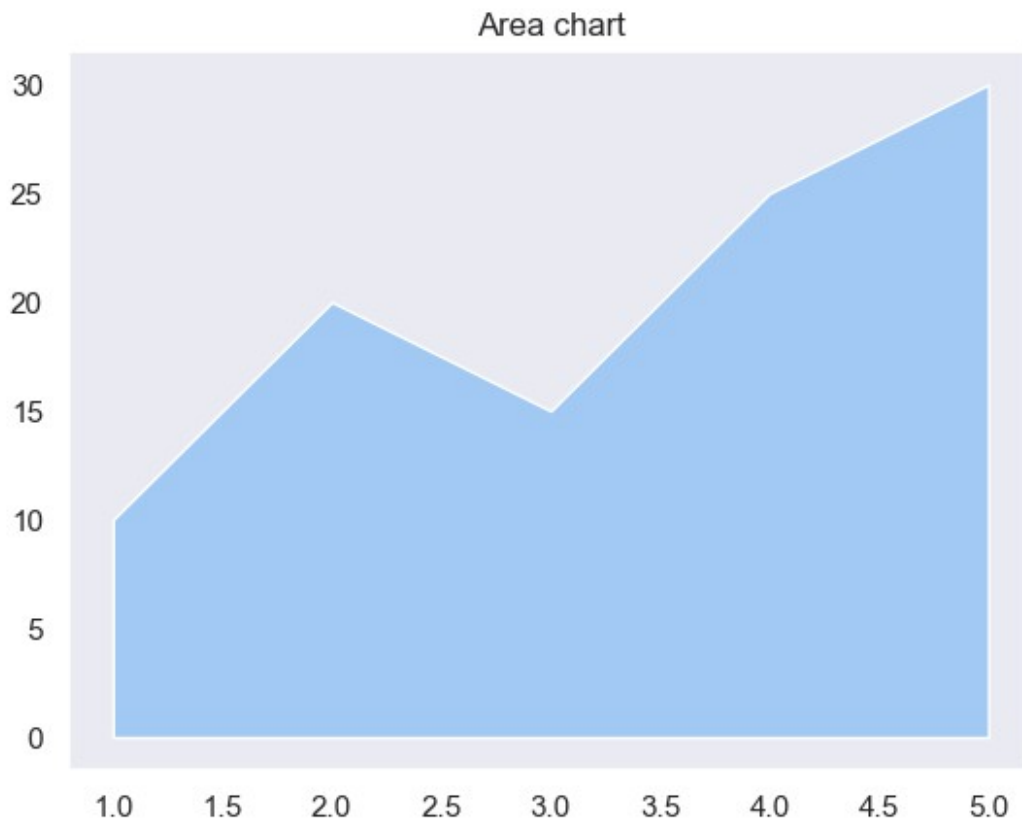
plt.scatter(hours, marks)

plt.title("Study Hours vs Marks")
plt.xlabel("Study Hours")
plt.ylabel("Marks")

plt.show()
```



```
import matplotlib.pyplot as plt  
  
x = [1,2,3,4,5]  
y = [10,20,15,25,30]  
  
plt.fill_between(x, y)  
  
plt.title("Area chart")  
plt.show()
```

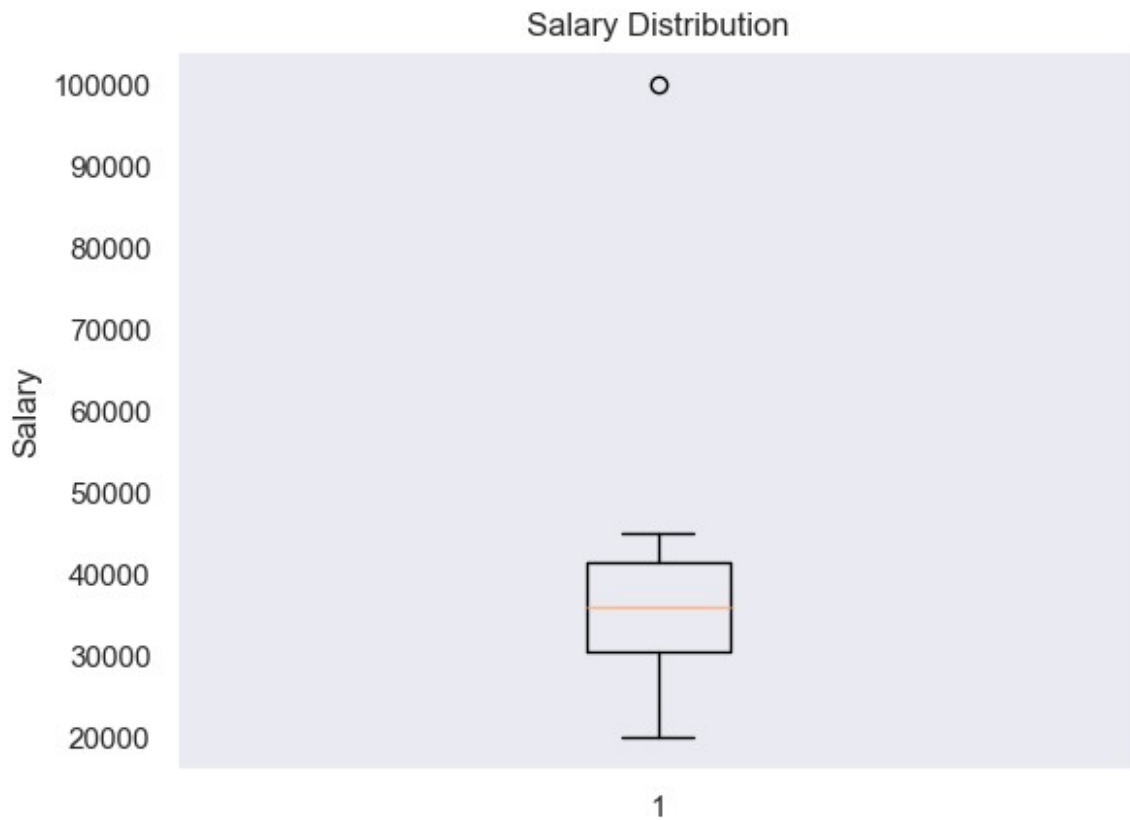


```
import matplotlib.pyplot as plt

salary = [25000, 20000, 30000, 32000, 35000,
          37000, 40000, 42000, 45000, 100000]

plt.boxplot(salary)

plt.title("Salary Distribution")
plt.ylabel("Salary")
plt.show()
```

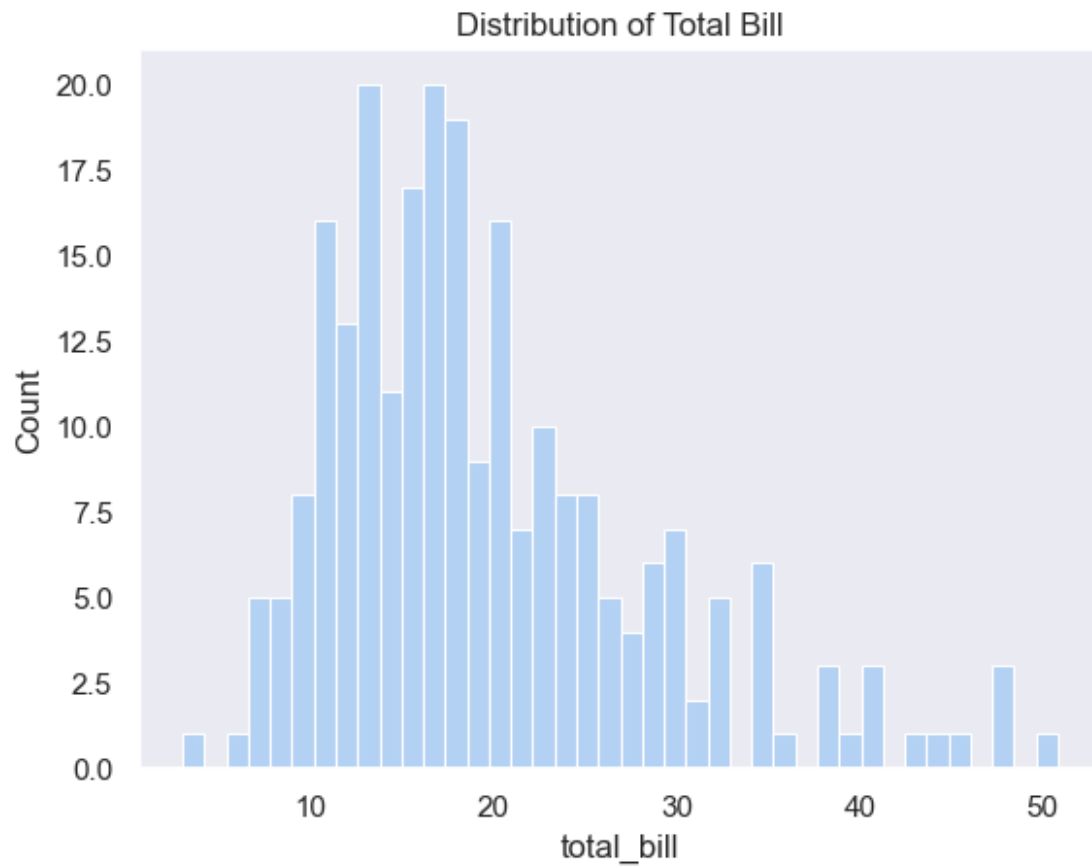


```
import seaborn as sns
import matplotlib.pyplot as plt
tips = sns.load_dataset("tips")

corr = tips.corr(numeric_only=True)

sns.histplot(
    data=tips,
    x="total_bill",
    bins=40
)

plt.title("Distribution of Total Bill")
plt.show()
```



```
sns.histplot(  
    data=tips,  
    x="total_bill",  
    kde=True  
)  
  
plt.title("Histogram with KDE")  
plt.show()
```

